25901751 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPHY100 - MODERN PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Give the postulates of special theory of relativity. [K] / [CO1]

   Ans. A. The laws of physics take the same form in all inertial frames of reference

   B. The speed of light in a vacuum is a universal constant, for all observers, regardless of their relative motion.

   Each postulate 1 mark.

2. What happens to the energy of a body as it approaches the speed of light? [A] / [CO1]

   Ans. Relativistic energy increases, approaching infinity as relativistic mass increases.

   Relation alone- 1 mark. Explanation/Relation with explanation -2 marks

3. Distinguish between inertial and non-inertial frames of reference. [U] / [CO1]

   Ans. A frame that is either at rest or moving at a constant velocity and in which Newton's first law holds true, are inertial frames of reference. A frame that is accelerating, decelerating, or rotating and in which Newton's first law doesn't hold true are non-inertial frames of reference.

   1 mark each for explaining one frame of reference. 2 marks for explaining both.

4. A red light has a wavelength of 632.8 nm. What is the frequency? [A] / [CO2]

   Ans. $\nu = \frac{c}{\lambda} = \frac{3X10^{8}}{632.8X10^{-9}} = 4.74X10^{14}\ Hz$

   1 mark for formula. Two marks for formula, substitution, calculation and result with unit.

5. How does the presence of a medium influence the propagation of electromagnetic waves? [U] / [CO2]

   Ans. Presence of medium mainly affects the speed of the electromagnetic wave. Speed in medium is always less than speed in vacuum.

6. Electron microscopes have better resolution than optical microscopes. Justify the statement. [A] / [CO2]

   Ans. Wavelength of electron beam is very small compared to optical wavelength. Hence better resolution.

   2 marks for correct answer.

7. In photoelectric effect blue light results in faster electrons than red light. Why? [U] / [CO2]

   Ans. $E = \phi + KE$; ϕ is the work function.

   For blue light E is large, thus KE is large for electrons emitted with blue light.

   Ans. 1 mark for equation alone. 2 marks for equation and explanation.

8. What are the features of a blackbody? [K] / [CO2]

   Ans. A. Ideal absorber which absorbs all radiation incident on it and reflects none.

   B. Emits a continuous range of radiation, which depends only on its temperature.

   1 mark each for one feature. 1 mark may be given if graph alone is plotted.

9. Explain the origin of emission and absorption spectral lines. [K] / [CO3]

Ans. Spectral lines are obtained when electron jumps between discrete energy levels in atoms. Emission spectral lines appear when electrons jump from higher energy state to lower, emitting energy. The result is bright or coloured lines against a dark background. Absorption spectral lines occur when electrons make a transition from lower to higher energy level which is possible only by absorbing energy. The result is dark lines in bright background because that energy has been absorbed.

Ans. 1 mark for each.

10. Explain the concept of atomic excitation in terms of electron transitions. [U] / [CO3]

Ans. Atomic excitation occurs when atomic electrons absorb energy from light, heat or collisions and jump from lower energy to higher energy state. This transition is quantized and occurs when the energy difference between energy levels is met by excitation sources.

2 marks.

11. Give the expansion of the acronym LASER. [K] / [CO4]

Ans. Light Amplification by Stimulated Emission of Radiation.

2 marks for correct answer.

12. Write the expressions for energy eigen value and wave function corresponding to the lowest energy level of a particle in a one dimensional box of length L [A] / [CO5]

Ans. $\psi = \sqrt{\frac{2}{L}} Sin \frac{n\pi x}{L}$

$E_n = \frac{n^2 h^2}{8mL^2}$

1 mark for each equation

13. Write steady state form of Schrodinger wave equation in one dimension. [K] / [CO5]

Ans. $\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$

2 marks for the correct equation.

14. What is a wave function in quantum mechanics? Explain its physical significance. [U] / [CO5]

Ans. A complex-valued function which varies with space and time, and which represents the state of the quantum system at any instant. As such it doesn't have any physical significance, but it is related to the probability of finding the particle.

1 mark for wavefunction and 1 mark for significance.

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

15. The measured life of a particle moving with a velocity 0.92 c is $3.4 \times 10^{-6}\,s$. Calculate its proper life. [A] / [CO1]

Ans. $Measured\ t = \frac{t_0}{\sqrt{1-\frac{v^2}{c^2}}}$ [2 marks]

$Proper\ life,\ t_0 = t.\sqrt{1-\frac{v^2}{c^2}} = 3.4X10^{-6}\sqrt{1-0.92^2} \simeq 1.3X\,10^{-6}\,s$ [3 marks for substitution, calculation and answer with unit]

16. Discuss the concept of length contraction. Obtain the relation between measured length and proper length. [U] / [CO1]

2 marks for explaining concept.

3 marks for obtaining $L = L_0\sqrt{1-\frac{v^2}{c^2}}$

17. A metal has a work function of 3 eV. What is the longest wavelength of light that can cause photoelectric emission from this metal? [A] / [CO2]

Ans. Work function $\phi = h\nu_0 = 3\ eV$

$$h\nu_0 = \frac{hc}{\lambda_0},\ \therefore \lambda_0 = \frac{hc}{\nu_0} \simeq \frac{1240\ eV.nm}{3\ eV} \simeq 413\ nm$$

[2 marks for equation of λ_0. 3 marks for substitution, calculation and answer with unit]

18. Find the de Broglie wavelength of a 1.0 mg grain of sand blown by the wind at a speed of 20 m/s. [A] / [CO2]

Ans. $\lambda = \frac{h}{p} = \frac{h}{mv}$ [2 marks]

$\lambda = \frac{h}{mv} = \frac{6.625X\,10^{-34}}{1.0X10^{-6}X20} \simeq 3.3X10^{-29}m$ [3 marks for substitution, calculation and answer with unit]

19. The position and momentum of a 1 KeV electron are simultaneously determined. If its position is located within 1 angstrom, what is the percentage of uncertainty in its momentum? [A] / [CO2]

Ans. $E = 1\ KeV = 1000X1.6X10^{-19} = 1.6X10^{-16}\,J$

$p = \sqrt{2mE} = \sqrt{2X9.1X10^{-31}X1.6X10^{-16}} \simeq 1.7X10^{-23}\,\frac{kgm}{s}$ [2 marks]

$\Delta p = \frac{\hbar}{2\Delta x} = \frac{1.054X10^{-34}}{2X1X10^{-10}} \approx 5.27X10^{-25} \frac{kgm}{s}$ [2 marks]

$Percentage\ uncertainity\ \frac{\Delta p}{p} X100 = \left(\frac{5.27X10^{-25}}{1.7X10^{-23}}\right) X100 \approx 3.1\ \%$ [1 mark]

20. Explain why the spacing between spectral lines decreases as we move towards higher quantum numbers in each hydrogen spectral series. [U] / [CO3]

Energy $E_n = \frac{-13.6\ eV}{n^2}$ [2 marks]

When $n = 1,\ 2,\ 3\ etc,\ E_1 = -13.6\ eV,\ E_2 = -3.4\ eV,\ E_3 = -1.51\ eV\ etc$

Hence the spacing between energy levels decreases with increase in n, hence the spacing between spectral lines. [3 marks for explanation]

21. Using a neat diagram, describe the process of stimulated absorption, spontaneous emission and stimulated emission of radiation. [U] / [CO4]

Ans. 2 marks each for describing stimulated absorption, 1 mark for spontaneous emission and 2 marks for stimulated emission

22. Using classical wave equation arrive at the time dependent Schrodinger wave equation in one dimension. [U] / [CO5]

Ans. Wave equation [2 marks]

Taking ψ, finding derivative with respect to $x\ and\ t$ and obtaining time dependent Schrodinger equation. [3 marks]

[5x6 = 30]